

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024	Date:	

Code of Conduct

I Vasanthan Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.
5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.

2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)
- d) Generate a simple English paraphrase without ambiguity.
- e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	4	4	3	3	18	94
2	5	3	4	4	3	19	
3	4	5	3	4	3	19	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1 — Reference Resolution and Discourse Coherence

Given paragraph:

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

a) Referring expressions and possible antecedents

Referring expression Possible antecedents

He	Ravi / Arun
him	Ravi / Arun
it	book / Artificial Intelligence
they	Ravi and Arun
the book	book introduced earlier

b) Apply the constraints

The assessment specifies **gender/number agreement, recency, grammatical role, semantic compatibility, and discourse coherence**.

Expression Antecedent		Valid?	Reason
He	Ravi	✓	Male, singular; grammatically possible
He	Arun	✓	Male, singular; also possible
him	Ravi	✓	Male, singular and semantically possible
him	Arun	✗	Arun is the current subject of “helped”; Ravi is the natural object/reference
it	book	✓	Singular, neuter; “find the book” is semantically valid
it	Artificial Intelligence	✗	“find Artificial Intelligence” does not fit the discourse meaning
they	Ravi + Arun	✓	Plural antecedent and both are active discourse entities

Important ambiguity

“He” is genuinely ambiguous between Ravi and Arun.

The given constraints do not completely eliminate either candidate:

He → Ravi ✓

He → Arun ✓

So the final resolution needs **discourse coherence/context**, not just gender or number.

c) Constraint Table

Referring Expression Possible Antecedents Valid/Invalid Main Constraint

He	Ravi	Valid	Agreement + coherence
He	Arun	Valid	Agreement + recency
him	Ravi	Valid	Grammar + semantic compatibility
him	Arun	Less preferred	Grammatical/discourse role
it	book	Valid	Semantic compatibility
it	Artificial Intelligence	Invalid	Semantic incompatibility
they	Ravi + Arun	Valid	Number + discourse coherence

The strongest resolution is therefore:

He → Ravi

him → Ravi

it → book

they → Ravi + Arun

d) Final Coreference Chains

Ravi

↓

He

↓

him

↓

they —————

|

Arun —————

book

↓

it

↓

the book

e) Rewritten Paragraph

Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

This removes the ambiguity while preserving the original discourse meaning.

f) If “He was looking for a book” appears before Arun is introduced

If the paragraph were:

“Ravi met at the library. He was looking for a book on Artificial Intelligence. Arun arrived later.”

then **He** → **Ravi** would be strongly preferred because Ravi is the only introduced male entity at that point.

But if **Arun had not yet been introduced at all**, the system cannot consider Arun as an antecedent. Once Arun is introduced later, it cannot retroactively resolve the earlier **He** to Arun without additional discourse evidence.

Conclusion

Reference resolution should combine **agreement, recency, grammatical role, semantic compatibility, and discourse coherence** rather than relying on a single rule. In this paragraph, **“it”** → **the book** and **“they”** → **Ravi + Arun** are clear, while **“He”** requires contextual/discourse reasoning. This constraint-based approach maintains consistent entity chains across the paragraph.

Q2 — Dialog Act Recognition and Response Generation

Given:

Student: “I submitted my assignment, but I am worried that I may have made many mistakes.”

a) Required Dialog Acts

The required dialog acts are:

- **Reassure** — reduce the student's worry.
 - **Advise** — provide constructive guidance.
-

b) Important Entities

The response should maintain:

- **assignment** — the submitted work
- **mistakes** — possible errors in the assignment
- **you** — the student

These entities should remain consistent throughout the response.

c) Three Possible Responses

Response 1

You have already submitted the assignment, so don't worry too much about possible mistakes. Review the feedback when you receive it and use it to improve your work next time.

Response 2

It is normal to feel worried after submitting an assignment, but you should remain confident. Review any feedback you receive and use it to improve areas where mistakes were made.

Response 3

You have completed the assignment, so stay confident and focus on what you can learn from it. Review the feedback carefully and use it to improve your future assignments.

All three are **2 sentences**, maintain the assignment/mistakes context, use a positive and polite tone, and include required keywords such as **review**, **improve**, **confident**, and **feedback**.

d) Comparison

	Response	Coherence	Constraint Satisfaction	Politeness	Relevance
1	High	High		High	High
2	High	High		High	High
3	Very High	Very High		Very High	Very High

e) Best Response

Response 3 is the best:

“You have completed the assignment, so stay confident and focus on what you can learn from it. Review the feedback carefully and use it to improve your future assignments.”

Justification

It satisfies all the constraints:

- **Reassurance:** “stay confident”
- **Advice:** “Review the feedback...”
- **Entity coherence:** assignment → feedback → future assignments
- **Discourse relation:** Cause–Effect/Elaboration
- **Keywords:** confident, review, feedback, improve
- **Length:** 2 sentences
- **Tone:** positive and polite
- **No contradictory advice**

Therefore, it provides the most coherent and constraint-satisfying response.

f) If Entity Coherence Is Not Maintained

If the system loses track of the entities, the response may become **irrelevant or contradictory**.

For example:

“You should review the mistakes in your **exam**.”

The original conversation is about an **assignment**, not an exam. This breaks entity coherence and makes the response less relevant.

Therefore, maintaining consistent references to **assignment, mistakes, and you** is essential for a coherent conversational response.

Q3 — Word Sense Disambiguation and Meaning Representation

Given:

“Priya sat on the bank and watched the boats moving across the water.”

a) Possible meanings of “bank”

1. **Financial institution**
 2. **Side of a river**
-

b) Correct Sense

The correct sense is:

Bank = side of a river

Justification

The surrounding context contains:

- **sat on the bank**
- **boats**
- **water**
- **moving across the water**

These are strong contextual and semantic cues for a **river bank**, not a financial institution.

Therefore:

bank → river bank

This satisfies the WSD constraint of using contextual information to select the intended meaning.

c) Predicate Logic Representation

Using the predicates suggested in the assessment:

sit(Priya, Bank)

bank(Bank)

beside(Priya, Bank)

watch(Priya, Boat)

boat(Boat)

move(Boat)

water(Water)

across(Boat, Water)

A more compact representation is:

sit(Priya, Bank) $\wedge$

bank(Bank) $\wedge$

beside(Priya, Bank) $\wedge$

watch(Priya, Boat) $\wedge$

boat(Boat) $\wedge$

move(Boat) $\wedge$

water(Water) $\wedge$

across(Boat, Water)

This preserves the relationships between **Priya, the river bank, boats, and water**.

d) Unambiguous Paraphrase

Priya sat on the side of the river and watched the boats moving across the water.

This explicitly identifies **bank** as the **side of a river**, removing the financial-institution interpretation.

e) How WSD Improves Machine Translation

WSD determines the intended meaning of an ambiguous word **before translation**.

For example:

English:

Priya sat on the bank.

WSD:

bank $\rightarrow$ river bank

$\downarrow$

Translation:

uses the target-language word for

the side of a river

Without WSD, a machine translation system could select the wrong sense of **bank**, producing an incorrect translation.

Therefore, WSD improves machine translation by **preserving the intended semantic meaning and relationships of ambiguous words**.